

Notes: Stavrakeva (RES, 2020)
Optimal Bank Regulation and Fiscal Capacity

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1 Big picture

- **Question.** Should countries with different fiscal capacity impose different ex ante bank regulation, especially different minimum bank capital requirements?
- **Core answer.** Yes. Countries with larger fiscal capacity should optimally have lower ex ante minimum bank capital ratios when capital requirements are used as the quantity instrument.
- **Key mechanism.** Larger fiscal capacity means that the government can provide a larger crisis bail-out at lower marginal social cost. This props up fire-sale prices and makes ex ante banking investment socially less costly.
- **Why regulation is needed.** The model has two externalities: inefficient pecuniary externalities from fire sales and, when banks are large, Too-Big-To-Fail moral hazard.
- **Economic takeaway.** Harmonized global bank-capital rules are theoretically problematic because the optimal rule depends on each country's ability to finance bail-outs and on whether the banking system is large enough to create TBTF moral hazard.

2 Incremental contribution of the paper

- **Fiscal capacity enters optimal bank regulation.** The paper brings government financing capacity into a theoretical banking-regulation model.
- **Quantity versus price regulation.** It shows that a capital requirement and a Pigouvian tax can implement the same allocation but move differently with fiscal capacity.
- **TBTF moral hazard is formalized.** Large banks internalize that larger crisis losses induce larger bail-outs, but not the cost of financing them.
- **Derivatives regulation becomes a direct implication.** Capital requirements alone may be insufficient if banks can write state-contingent claims against crisis states.
- **Fiscal capacity is decomposed.** Fiscal capacity from a large tax base and fiscal capacity from efficient tax collection can have different implications for moral hazard.

Interpretation: The incremental contribution is not merely saying that bail-outs matter. It shows that the government's ability to finance bail-outs changes both the optimal capital ratio and the number of instruments needed.

3 Data and measurement / model primitives

The paper is a theoretical model and therefore has no empirical estimation sample. The relevant “measurement” objects are model primitives: fiscal capacity, bank investment, fire-sale prices, bail-outs, and regulatory instruments.

3.1 Time, states, and agents

- There are three periods, $t = 0, 1, 2$.
- At $t = 1$, the economy can be in a good state g or a bad crisis state b .
- Consumers supply labour and finance banks using state-contingent contracts.
- Bankers choose risky investment and state-contingent borrowing.

- Foreign arbitrageurs buy capital in fire sales and have lower productivity than bankers.
- A bail-out authority taxes consumers and transfers resources to banks during a crisis.
- A constrained central planner provides the benchmark allocation.

Interpretation: The bad state is where fire sales and bail-outs become relevant.

3.2 Consumers and fiscal resources

$$U = c_0^c - \omega l_0 + \sum_{t=1}^2 \sum_s \pi_s (c_{t,s}^c - \omega l_{t,s}), \quad y_{t,s} = a l_{t,s}^\alpha, \quad 0 < \alpha < 1.$$

$$l_{t,s}^*(\tau_{t,s}) = \left(\frac{(1 - \tau_{t,s}) \alpha a}{\omega} \right)^{1/(1-\alpha)}.$$

Interpretation: Distortionary taxation lowers labour supply and output. Fiscal capacity comes from being able to raise bail-out funds at lower marginal social cost.

3.3 Bail-out technology and fiscal capacity

$$T_{t,s} \leq \chi \tau_{t,s} a l_{t,s}^\alpha, \quad 0 \leq \chi \leq 1.$$

- T is the transfer to banks.
- τ is the labour tax rate.
- χ measures tax-collection efficiency and the lack of corruption or leakage.
- a and ω shape the tax base through productivity and labour disutility.

Fiscal capacity = lower marginal cost of one extra dollar of bail-out for a fixed bail-out level.

Interpretation: Fiscal capacity is not just GDP. It includes the size of the tax base and the efficiency with which taxes become transfers.

3.4 Banks and regulatory instruments

$$k_0^i \leq \frac{n_0^i}{\rho^i}, \quad q_0(1 + \tau_0^{k,i}), \quad d_{1,b}^i \leq \nu^i.$$

Interpretation: The first expression is a capital requirement. The second is an investment tax. The third is a cap on bad-state liabilities, the model analogue of restricting CDS-like exposures.

4 Models and theoretical specifications

4.1 Fire sales and pecuniary externalities

In a crisis, banks sell capital to foreign arbitrageurs. Because arbitrageurs are less productive users of capital, this fire sale is socially costly. Small banks ignore how their own investment contributes to aggregate fire-sale pressure, creating overinvestment.

Interpretation: The inefficient price effect is that private choices move equilibrium prices and tighten others' constraints, but each small bank ignores this impact.

4.2 Small banks: one instrument is enough

With infinitesimal banks, there is no TBTF moral hazard. The only key distortion is ex ante overinvestment from fire-sale pecuniary externalities. Therefore one instrument controlling investment can decentralize the constrained planner allocation:

$$\rho^* = \frac{n_0}{k_0^{CP}}, \quad \tau_0^{k,*} = \text{inefficiency wedge at } k_{1,b}^{f,CP}.$$

Interpretation: Fixing the investment quantity fixes the distortion in the small-bank benchmark.

4.3 Fiscal capacity and capital requirements

$$\frac{\partial \rho^*}{\partial \chi} < 0, \quad \frac{\partial \rho^*}{\partial a} < 0, \quad \frac{\partial \rho^*}{\partial \omega} > 0.$$

Interpretation: With a quantity instrument, a high-fiscal-capacity country can allow higher investment because it can cushion crisis fire sales. Since ρ is inversely related to investment, higher fiscal capacity implies a lower required capital ratio.

4.4 Large banks and TBTF moral hazard

Large banks internalize that larger fire sales imply larger bail-outs, but they do not internalize the fiscal cost of those bail-outs. This creates TBTF moral hazard and can make them overborrow against the bad state.

Interpretation: A capital requirement controls asset size but may not control state-contingent liabilities. This is why a second instrument may be needed.

4.5 Large-bank policy instruments

$$\begin{aligned} \frac{\partial \rho^{l,*}}{\partial \chi} &< 0, & \frac{\partial \rho^{l,*}}{\partial a} &< 0, & \frac{\partial \rho^{l,*}}{\partial \omega} &> 0. \\ \frac{\partial \tau_0^{k,l,*}}{\partial a} &> 0, & \frac{\partial \tau_0^{k,l,*}}{\partial \omega} &< 0, & \frac{\partial \tau_0^{k,l,*}}{\partial \chi} &< 0. \\ \frac{\partial \nu^{l,*}}{\partial \chi} &> 0, & \frac{\partial \nu^{l,*}}{\partial a} &> 0, & \frac{\partial \nu^{l,*}}{\partial \omega} &< 0. \end{aligned}$$

Interpretation: The signs differ because quantity instruments are tied to the constrained planner's desired quantity, while price instruments are tied to the externality wedge.

5 Identification and theoretical design

Because the paper is theoretical, this section is about model identification rather than empirical causal identification.

5.1 Objects being compared

- The decentralized equilibrium without regulation gives private bank choices.
- The constrained central planner allocation gives the welfare benchmark subject to the same market incompleteness and fiscal constraints.
- The gap identifies the externalities.
- The policy maker’s problem identifies which instruments replicate the planner allocation.

Interpretation: The paper identifies distortions by comparing allocations inside the model, not by estimating regressions.

5.2 Small versus large banks

The contrast between small and large banks isolates TBTF moral hazard. Small banks generate pecuniary externalities but take bail-outs as given. Large banks understand that their own crisis losses increase the aggregate bail-out.

Interpretation: Adding strategic bank size creates the need for the crisis-state liability cap.

5.3 Fiscal-capacity comparative statics

The paper varies a , ω , and χ . All three can lower the marginal cost of bail-outs, but they do not affect TBTF moral hazard identically.

Interpretation: The source of fiscal capacity matters: a large tax base can strengthen TBTF incentives, while efficient tax collection can weaken them.

5.4 Instrument distinction

Capital requirements implement a quantity. Investment taxes price an externality wedge. This is why larger fiscal capacity can imply lower capital requirements but higher Pigouvian taxes in some large-bank cases.

Interpretation: The phrase “more regulation” is ambiguous unless the regulatory instrument is specified.

5.5 Limits of the theoretical design

The model abstracts from sovereign borrowing, monetary policy, deposit insurance, detailed bank balance sheets, cross-border relocation, and regulatory arbitrage. These simplifications limit direct implementation but clarify the core mechanism.

6 Figures

The paper has no empirical tables in the main text. Its only original figure is an appendix figure documenting cross-country dispersion in minimum required risk-based regulatory capital ratios.

6.1 Figure A1: Minimum Required Risk-Based Regulatory Capital Ratio – Density

Read: Most countries cluster around the Basel-style 8% minimum, but some have ratios near 10–12%, 15–16%, and above 18%. The figure documents partial harmonization with some cross-country dispersion.

Economic message: The figure motivates the paper’s puzzle: capital regulation is partly harmonized internationally, but bail-out capacity differs across countries.

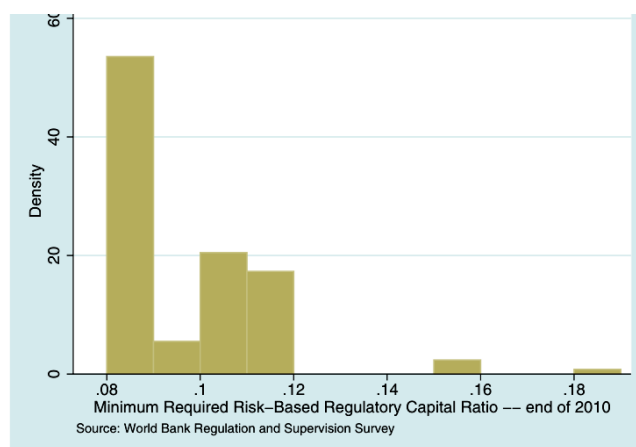


FIGURE A1
Minimum Required Risk-Based Regulatory Capital Ratio – Density.

Figure A1: Minimum Required Risk-Based Regulatory Capital Ratio – Density. The histogram illustrates that many countries use similar Basel-style minimum capital ratios even though their fiscal capacity differs. The model asks whether such harmonization is normatively justified.

6.2 Result box: small banks and capital requirements

Read: Small banks overinvest because they do not internalize fire-sale price effects. A single ex ante instrument can implement the constrained planner allocation.

Economic message: A fiscally strong country can optimally tolerate more bank investment and therefore impose a lower capital requirement.

6.3 Result box: large banks and derivatives regulation

Read: Large banks may use state-contingent liabilities to shift losses into crisis states. If TBTF moral hazard is strong, capital requirements alone are insufficient.

Economic message: The model gives a rationale for regulating CDS-like contracts even without counterparty risk.

7 Short summary

- The paper studies optimal bank regulation when countries differ in fiscal capacity.

- Fiscal capacity is the marginal cost of financing an extra dollar of bail-out.
- Fire sales create pecuniary externalities and ex ante overinvestment.
- With small banks, one instrument controlling ex ante investment is enough.
- With large banks, TBTF moral hazard may require a limit on bad-state liabilities.
- Larger fiscal capacity implies lower optimal minimum capital ratios when capital requirements are used.
- Price and quantity instruments can have different comparative statics.

8 Conclusion

8.1 Core conclusion

Fiscal capacity should be a central input into bank regulation. Countries that can finance crisis bail-outs at lower marginal social cost can optimally tolerate more ex ante bank investment and lower capital requirements.

8.2 Economic intuition

A fiscally strong government can support banks during a crisis, reducing fire-sale losses. A fiscally weak government cannot cushion fire sales as much, so it should require banks to carry more capital ex ante.

8.3 Incremental contribution

The paper adds the state's crisis-financing ability to optimal bank-regulation theory and shows that capital ratios, investment taxes, and derivatives constraints respond differently to fiscal capacity.

8.4 Policy implications

- Uniform global capital regulation may be inefficient.
- Capital requirements alone may be insufficient for large banks.
- Regulators should monitor banks' net liabilities in crisis states.
- CDS-like exposures can be important even absent counterparty-risk concerns.

8.5 Limitations and future research

Important extensions include sovereign borrowing, monetary policy, regulatory arbitrage, bank heterogeneity, deposit insurance, and richer bank balance sheets.

8.6 Takeaway

Optimal bank regulation is not only about banks. It is also about the state's ability to finance bail-outs in crisis states.